# Corrections to "Arbitrary-Direction SAR Ship Detection Method for Multiscale Imbalance"

Zhongzhen Sun, Xiangguang Leng, *Member, IEEE*, Xianghui Zhang, Zheng Zhou, Boli Xiong, Kefeng Ji, *Member, IEEE*, and Gangyao Kuang, *Senior Member, IEEE*

IN THE above article [1], the references in Table IX were incorrect. The corrected table is given here.

TABLE IX

COMPARISON OF MODEL COMPLEXITY AND COMPUTATIONAL EFFICIENCY WITH DIFFERENT ARBITRARY-DIRECTION SAR SHIP DETECTION ALGORITHMS ON R-SSDD

| Type | Method | Para(M) | FLOPs(G) | Weights(Mb) | FPS |
|---|---|---|---|---|---|
| Anchor-free | CFA [54] | 36.60 | 48.56 | 280.51 | 29.46 |
| | Oriented RepPoints [55] | 36.60 | 48.56 | 280.51 | 28.55 |
| | Rotated RepPoints [56] | 36.60 | 48.56 | 280.51 | 31.45 |
| | SASM RepPoints [57] | 36.60 | 48.56 | 280.51 | 31.38 |
| Two-Stage | Faster-RCNN-OBB [18] | 41.12 | 50.36 | 314.99 | 34.51 |
| | Gliding Vertex [58] | 41.13 | 50.37 | 315.02 | 33.83 |
| | Oriented R-CNN [59] | 41.13 | 50.40 | 315.00 | 30.19 |
| | ReDet [60] | 31.54 | 15.11 | 245.01 | 18.47 |
| | RoI Transformer [61] | 55.03 | 51.38 | 421.06 | 26.43 |
| One-Stage | ATSS-OBB [62] | 36.01 | 51.79 | 276.01 | 37.17 |
| | ConvNeXt [63] | 37.14 | 53.38 | 425.29 | 37.75 |
| | CSL [64] | 37.06 | 57.49 | 283.97 | 33.81 |
| | FCOS-OBB [65] | 31.89 | 51.55 | 244.55 | 38.69 |
| | GWD [66] | 36.13 | 52.39 | 276.85 | 36.92 |
| | KFIoU [67] | 41.58 | 82.17 | 318.50 | 26.84 |
| | KLD [68] | 39.22 | 69.29 | 300.48 | 29.58 |
| | PSC [69] | 36.25 | 53.07 | 278.71 | 30.78 |
| | R3Det [70] | 41.58 | 82.17 | 282.47 | 18.77 |
| | RetinaNet-OBB [71] | 36.13 | 52.39 | 276.85 | 36.22 |
| | RTMDet-OBB [49] | **8.86** | **9.40** | **141.26** | **139.28** |
| | S2ANet [72] | 38.54 | 49.05 | 295.25 | 29.86 |
| | **MSDFF-Net(Ours)** | 8.94 | 9.57 | 143.46 | 135.79 |

## REFERENCES

[1] Z. Sun et al., "Arbitrary-direction SAR ship detection method for multiscale imbalance," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5208921, doi: 10.1109/TGRS.2025.3559701.

Received 8 May 2025; accepted 8 May 2025. Date of current version 23 May 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62001480, Grant 62471475, and Grant 62401584; in part by Hunan Provincial Natural Science Foundation of China under Grant 2024JJ4045; in part by the Science and Technology Innovation Program of Hunan Province under Grant 2024RC3124; and in part by the Young Elite Scientists Sponsorship Program. *(Corresponding author: Xiangguang Leng.)*

The authors are with the State Key Laboratory of Complex Electromagnetic Environment Effects on Electronics and Information System, National University of Defense Technology, Changsha 410073, China (e-mail: sunzhongzhen14@163.com; luckight@163.com; zxhzxh0122@163.com; zhouzheng19961123@163.com; blxiong@nudt.edu.cn; jikefeng@nudt.edu.cn; kuangyeats@hotmail.com).

Digital Object Identifier 10.1109/TGRS.2025.3568914

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.